

Data Structures and Algorithms

A data structure is a special format for organizing, processing, retrieving and storing data so operations on it can be done effectively. An algorithm is a step-by-step procedure that transforms inputs into outputs, and it must be unambiguous, have defined inputs and outputs, terminate, and be feasible to implement. In short, data structures organize data, and algorithms process it. Common algorithm operations tied to data structures include search, sort, insert, update, and delete.

Linear Data Structure

Linear data structures arrange elements sequentially, so each element has a previous and next neighbor, except the first and last. The common types are arrays, linked lists, stacks (LIFO), and queues (FIFO). Because elements are connected in sequence, the structure can be traversed in a single run.

One-Dimensional Array

An array is a collection of items of the same data type stored at contiguous memory locations. A one-dimensional array is the simplest form, a single row of elements accessed by one index. Access time is $O(1)$ because elements sit in contiguous locations, while search, insertion, and deletion are $O(n)$ in the worst case.

The Big O Notation

Big O expresses an upper bound on an algorithm's time or space complexity, describing worst-case performance as a function of input size n . It only describes asymptotic behavior, not exact values, and is used to compare the efficiency of algorithms. The two things measured are time complexity (how long it runs) and space complexity (how much extra space it needs). The letter O comes from Ordnung, meaning order of approximation.

Linear Search

You start at the first position and compare each item to the target. If it matches, you return that position; if you reach the end without a match, you return -1. It's also called sequential search, works on unsorted data, and runs in $O(n)$.

Binary Search

Binary search is more optimized: it compares the target with the middle element, discards the half that can't contain it, and updates the lower and upper bounds. This gives $O(\log n)$ time, but the data must be sorted. It's also called half-interval search.

References

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